Memo - Measurement Revision

13.2 A rectangular prism has sides $3a$, $2a$, and a centimetres. The total surface area of the prism is $550cm^2$.

13.2.1 Determine the value of a. (4)

13.2.2 If all sides are multiplied by a factor of k the total surface area reduces to $137{,}5cm^2$. Determine the value of k. (4)

13.2.1

$TSA = 2(2a \times a) + 2(3a \times 2a) + 2(3a \times a)$

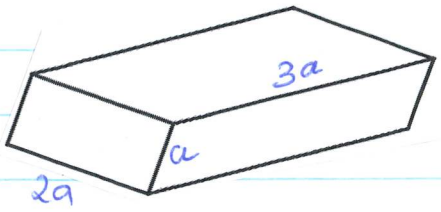
$= 4a^2 + 12a^2 + 6a^2$

$= 22a^2$

$550 = 22a^2$

$25 = a^2$

$5 = a$



13.2.2. Sides are 10, 5 and 15 (sub $a=5$)

[Multiplying each side by $k \rightarrow 10k ; 5k ; 15k$]

$TSA = 2(5k \times 10k) + 2(15k \times 10k) + 2(15k \times 5k)$

$= 100k^2 + 300k^2 + 150k^2$

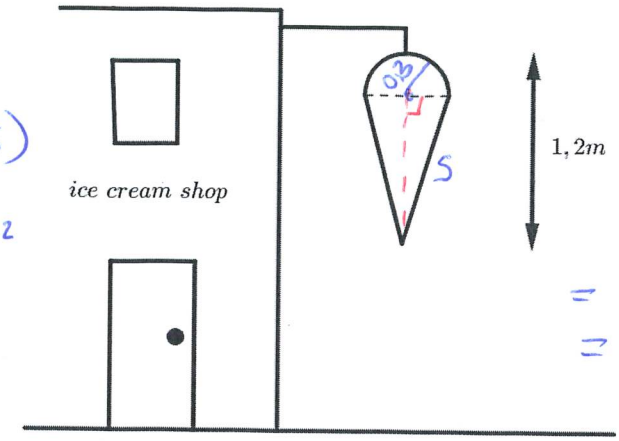
$= 550k^2$

$\therefore 137{,}5 = 550k^2$

$\frac{1}{4} = k^2$

$\frac{1}{2} = k$

13.4 An ice cream shop displays a large metal ice cream advertising its business. The shape is hollow and made up of a cone and a hemisphere as shown. The vertical height of the display from top to bottom is 1,2 metres and the radius of the hemisphere is 0,3 metres.



Slant ht (s) of cone

$s^2 = (0,3)^2 + 0,9^2$

$s^2 = \frac{9}{10}$

$s = \frac{3\sqrt{10}}{10}$

ht of cone

$= 1,2 - 0,3$

$= 0,9m$

13.4.1 Show that the total volume of the metal ice cream is $\frac{9}{200}\pi$ cubic metres. (4)

13.4.2 Calculate the total surface area of the metal ice cream. (4)

Useful formulae: $V_S = \frac{4}{3}\pi r^3$, $SA_S = 4\pi r^2$, $V_C = \frac{1}{3}\pi r^2 h$, $SA_C(open) = \pi rs$

13.4.1. Volume = cone + hemisphere

$= \frac{1}{3}\pi r^2 h + \frac{1}{2}\left(\frac{4}{3}\pi r^3\right)$

$= \frac{1}{3}\pi (0,3)^2 0,9 + \frac{1}{2} \times \frac{4}{3}\pi (0,3)^3$

$= \frac{27}{1000}\pi + \frac{9}{500}\pi$

$= \frac{9}{200}\pi\ m^3$ OR $0,14\ m^3$

13.4.2 TSA = cone + hemisphere

$= \pi rs + \frac{1}{2}(4\pi r^2)$

$= \pi (0,3)\left(\frac{3\sqrt{10}}{10}\right) + 2\pi (0,3)^2$

$= 1,46\ m^2$